

What is this?

This notebook is part of a collection of `pluto` notebooks on various topics discussed during the Time Domain Astrophysics course delivered by Stefano Covino at the Università dell'Insubria in Como (Italy). Please direct questions and suggestions to stefano.covino@inaf.it.

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TIME-DOMAIN ASTROPHYSICS

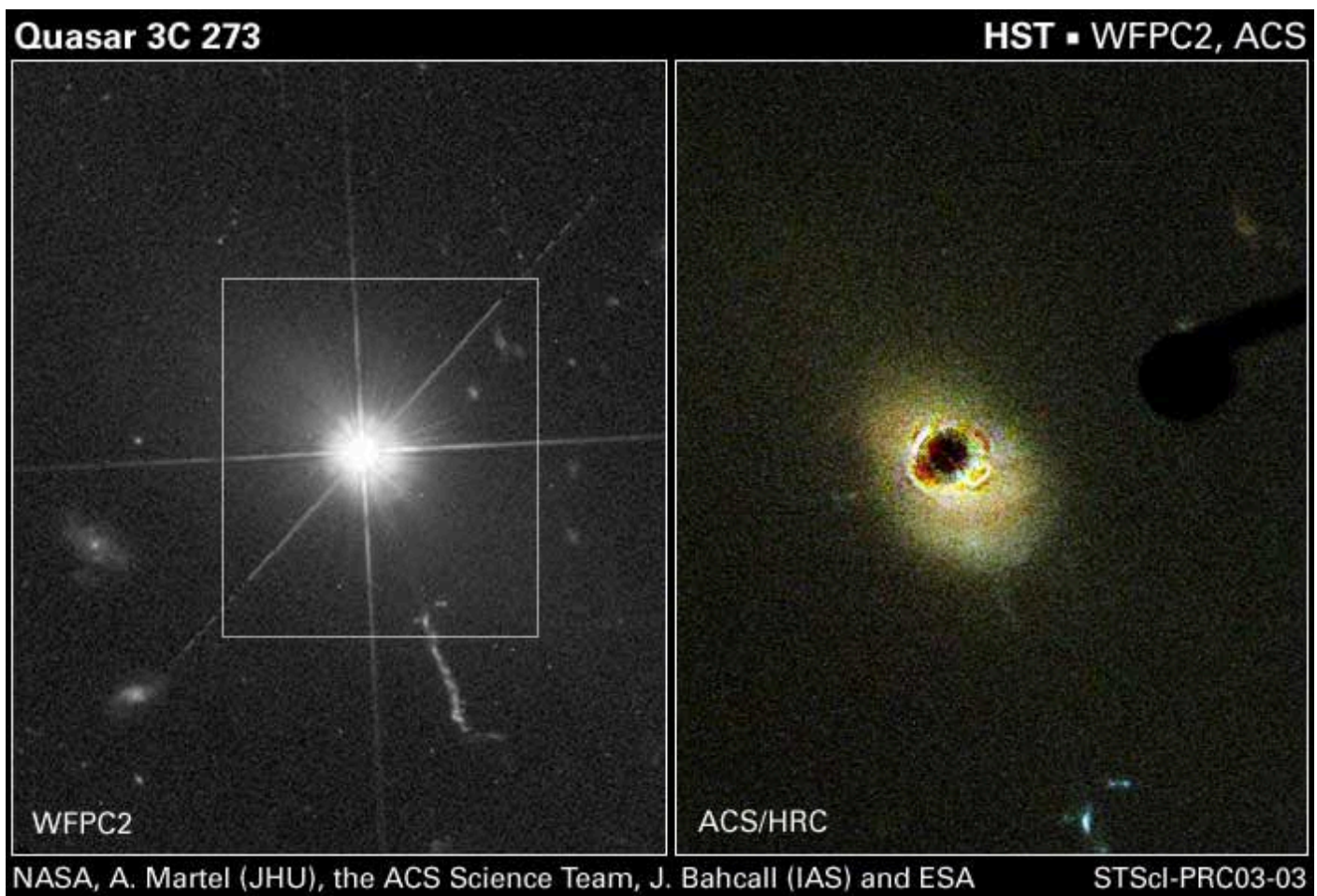
Stefano Covino

INAF / Brera Astronomical Observatory



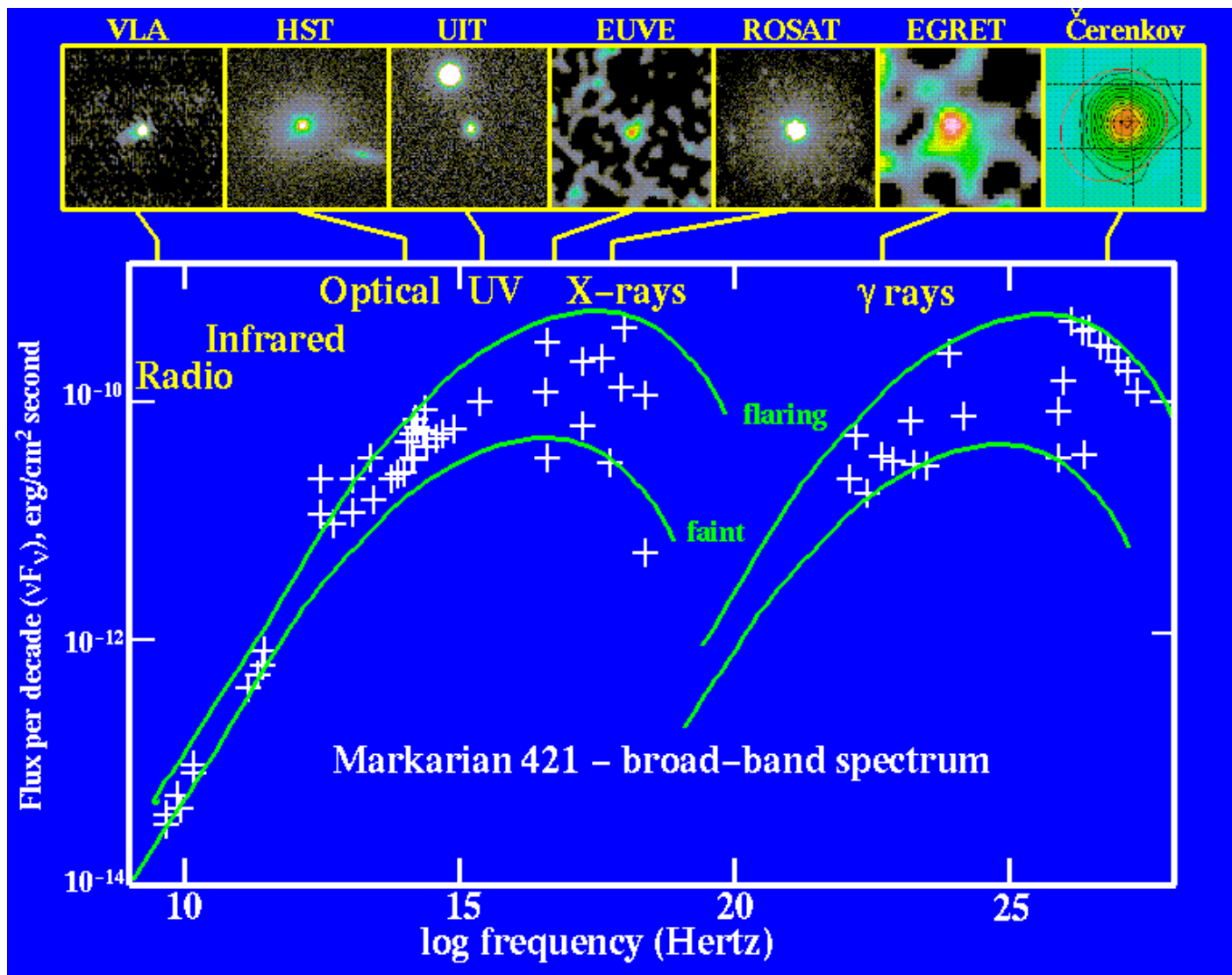
Science Case: AGN and Blazars

- Why are they special?
- What is the power source?
- What is the sources of continuum emission?
- What is producing emission and absorption features?
- How do jets form?
- AGN classification schema and unification.
- The co-evolution of black holes and galaxies.
- More specifically:
 - Very large luminosities are possible, up to 10,000 times a typical galaxy.
 - The emission spans a huge range of photon energy, i.e. from radio to gamma-rays.
 - The source of energy generation is very compact, typically lower than the size of the solar system.
 - In some cases, there is significant energy transported in relativistic jets.
- This AGN, 3C273, is several hundred times brighter than its host galaxy, just in visible light alone:

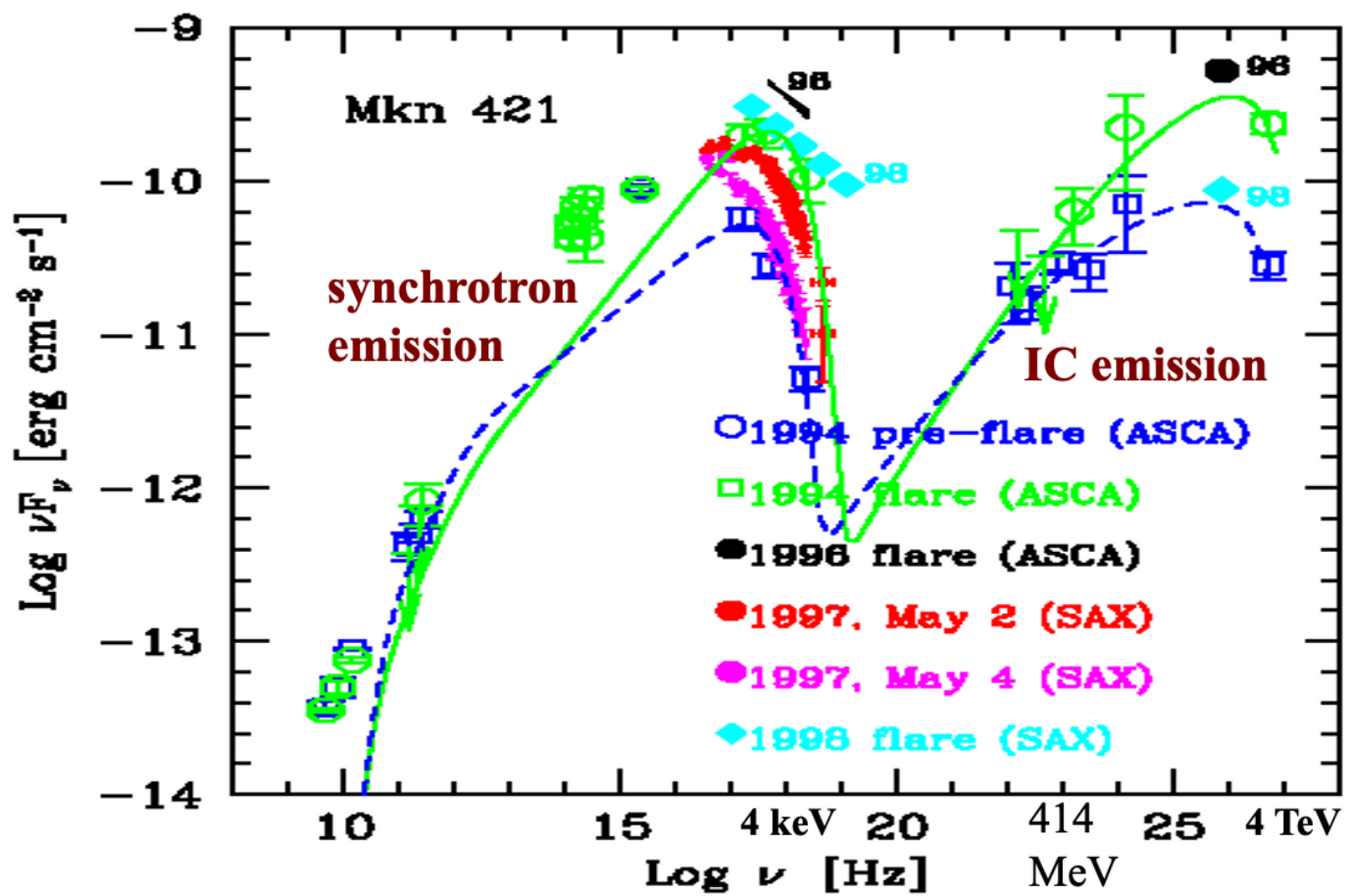


AGN: Observational data

- Highly luminous: $L_{\text{bol}} \sim 10^{42} - 10^{48} \text{ erg s}^{-1}$ ($10^9 - 10^{15} L_{\odot}$).
- Compact: size $\ll 1 \text{ pc}$.
- Broad-band continuum emission:
 - $dL/d\log f = \text{const.}$ From IR to X-rays and γ -rays
- Variable: on different times scales. Rapid variability in the X-ray indicate that emission comes from the innermost region.
- Strong Radio emitters: in some sources extended jets are present.
- Polarized emission.
- Markarian 421 shows comparable power emitted across about seven orders of magnitude in photon energy:



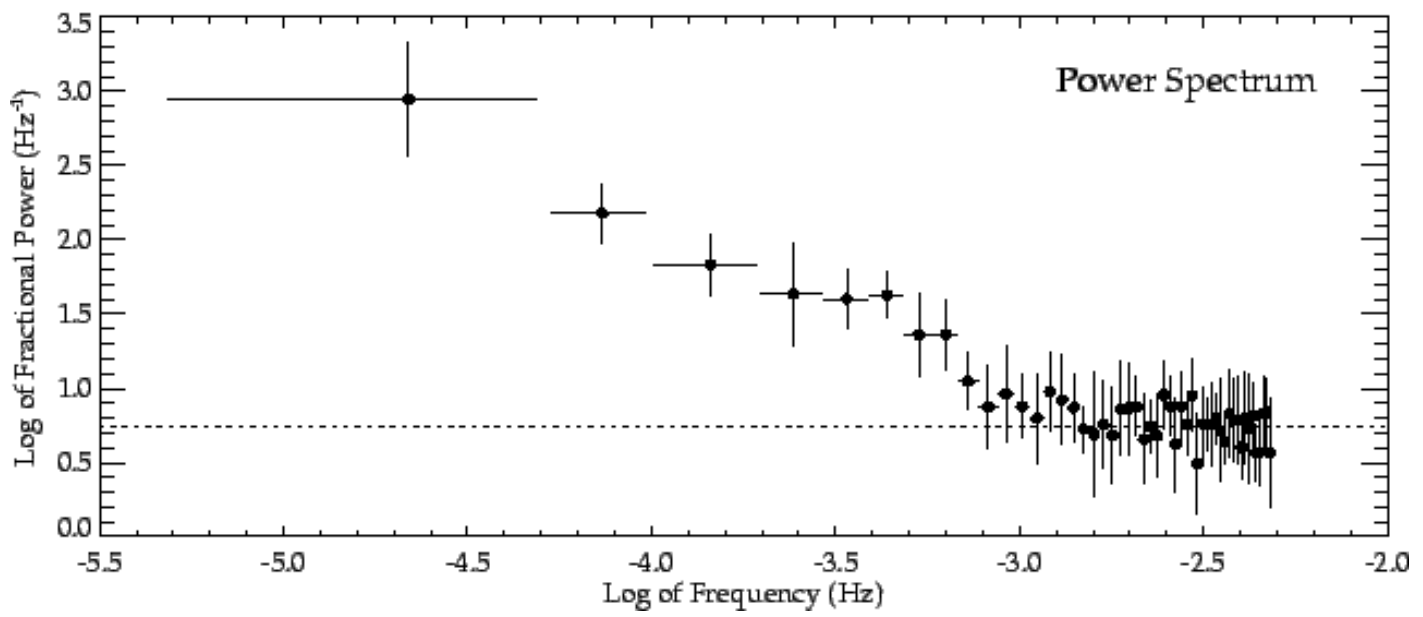
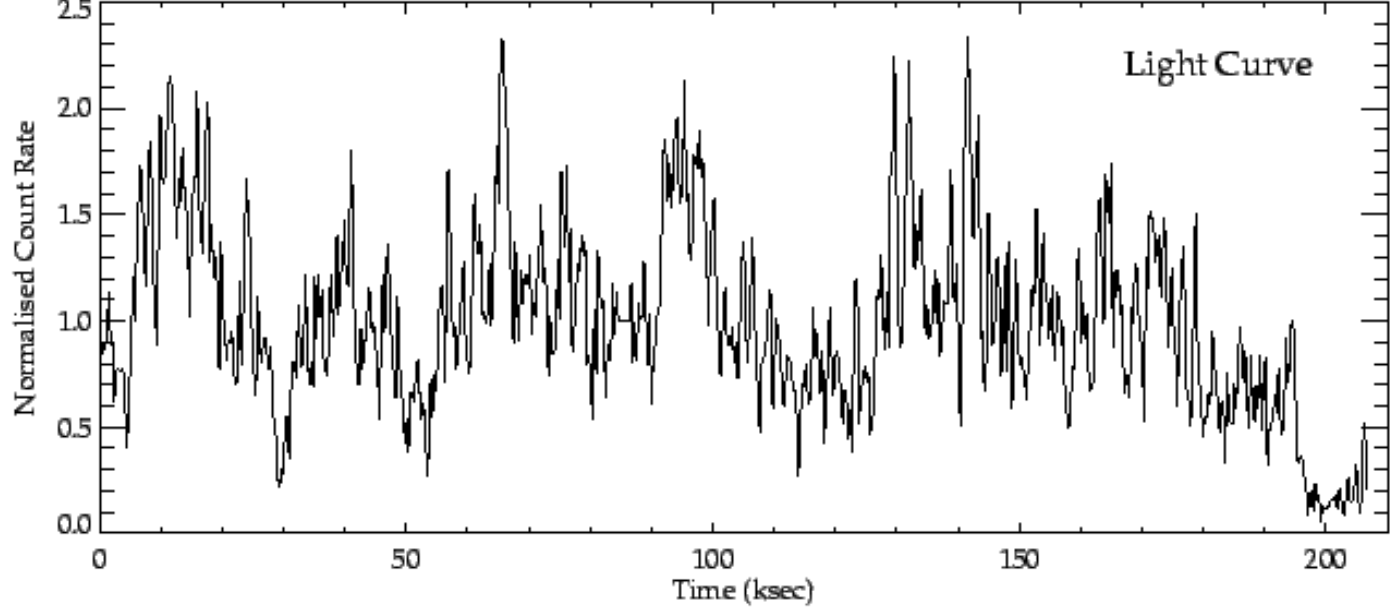
- Markarian 421 is also bright at TeV energies, nowadays accessible, e.g., by Cherenkov telescopes:



Tip

A simple, yet intriguing argument:

- A source that varies significantly in time t must have size $R < ct$, where c is the speed of light.



Exercise: compute the upper limit to the source size for Mrk 421:

- At ~ 100 keV the shortest observed variability is ~ 2.9 hours.
- The size of the emitting region turns out to be:

$$l \leq \frac{c\Delta t\delta}{1+z}$$

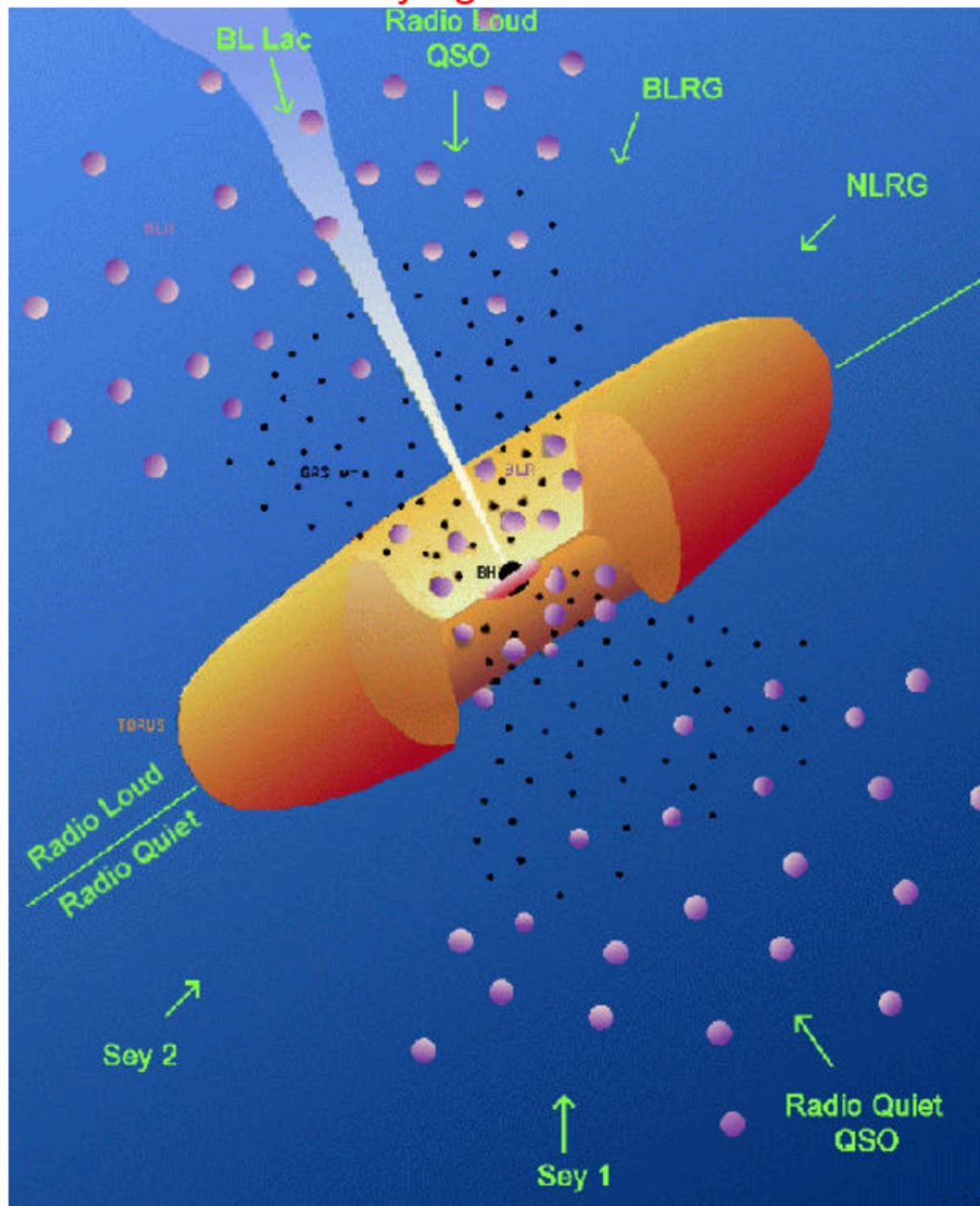
- where c is the speed of light, Δt the variability time, δ the Doppler factor and z the source redshift.
- The Doppler factor is $\delta = \frac{1}{\Gamma(1-\beta\cos\theta)} = \frac{\sqrt{1-\beta^2}}{1-\beta\cos\theta}$, where Γ is the Lorentz factor, β the source speed in units of the speed of light, and θ the angle of motion wrt the line of sight.
 - For this source: $z = 0.030$, $\theta \approx 0^\circ$ and $\beta > 0.98$ so that $\delta \sim 10$ and, finally, $l \leq 3 \times 10^{15}$ cm = 203 AU.

The AGN unified model

- Many of the various phenomenologies shown by AGN can be explained by a unique simple scenario and different viewing angles:

Unified Model

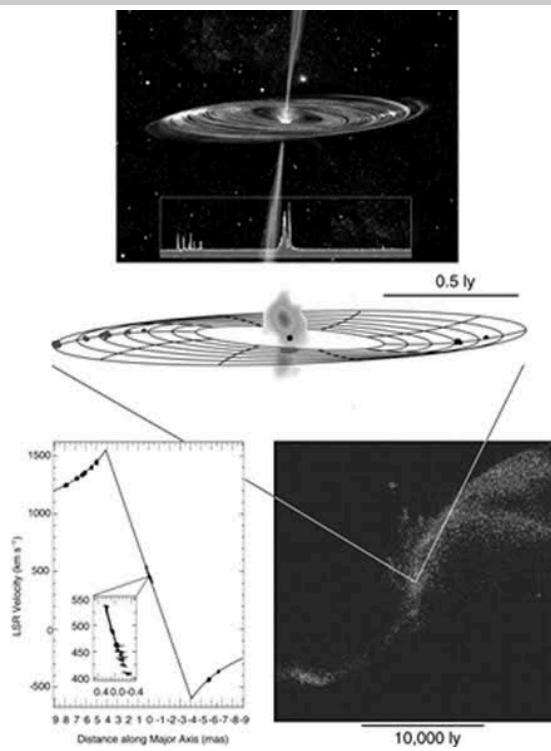
The Unifying Model of AGN



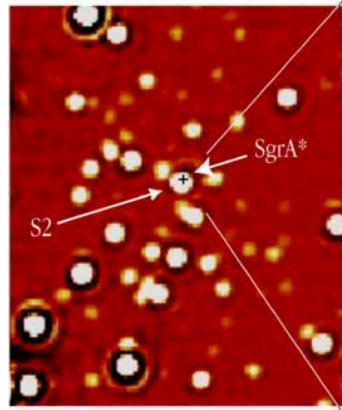
Urry & Padovani

IAAT

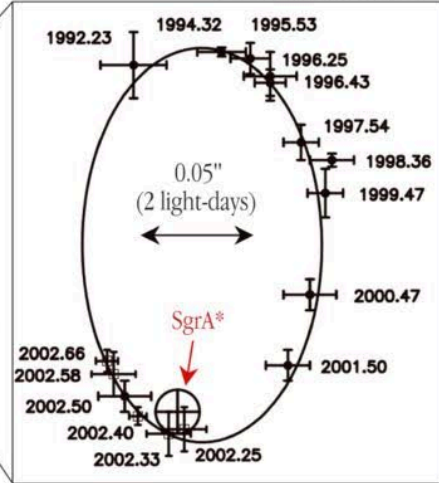
- The power source is accretion onto a supermassive black-hole.
 - Efficient, compact, and capable of producing high-energy emission and jets.
- It is occasionally possible to measure directly the mass of the central black-hole, e.g., by Newtonian dynamics ($M = v^2 R / G$):
 - Water masers have been mapped in NCC 4258: $M \sim 40 \times 10^6 M_{\odot}$.
 - Orbits of stars in the Galactic Center: $M \sim 3 \times 10^6 M_{\odot}$.



NACO May 2002



S2 Orbit around SgrA*



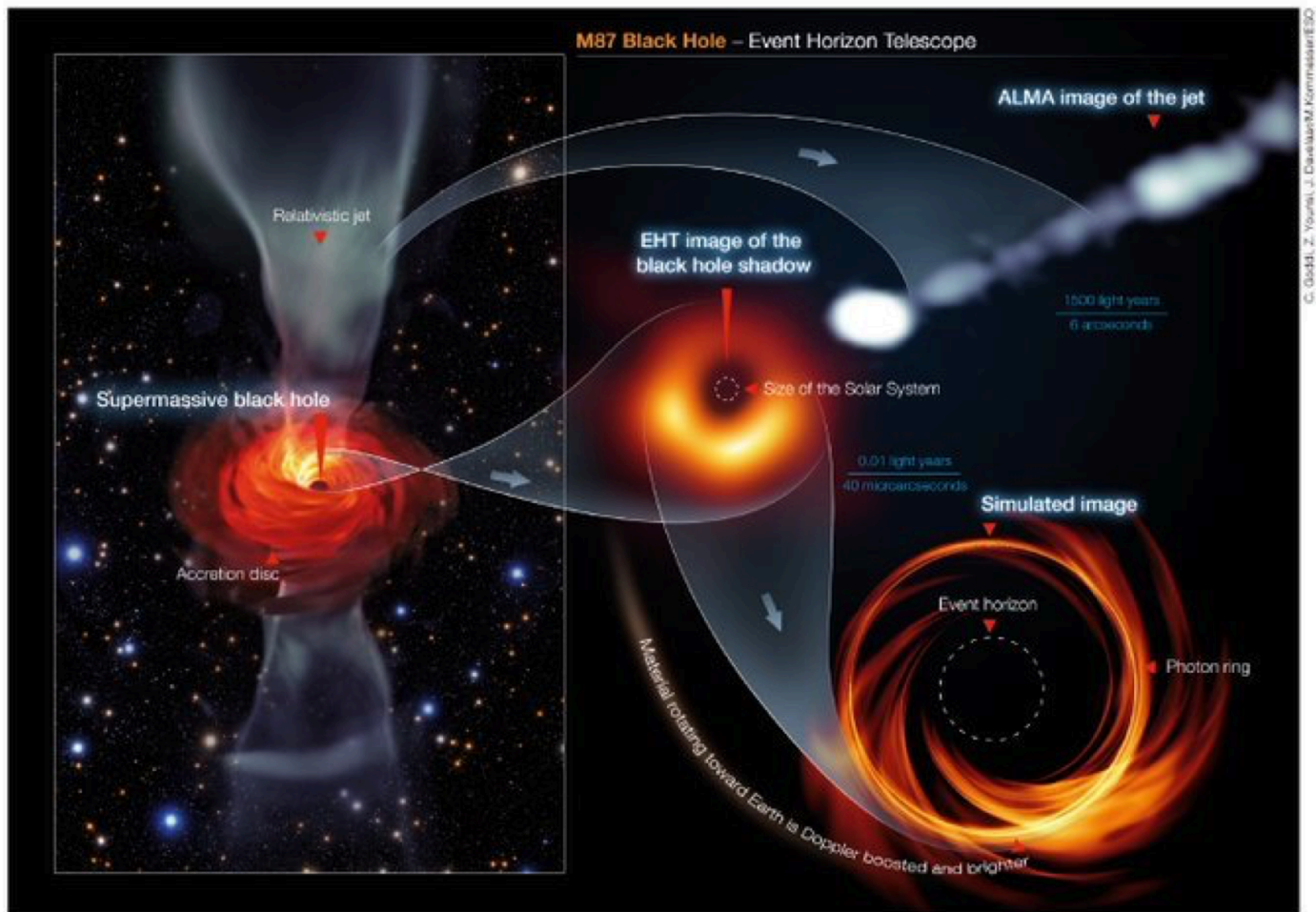
The Motion of a Star around the Central Black Hole in the Milky Way

ESO PR Photo 25c/02 (9 October 2002)

© European Southern Observatory



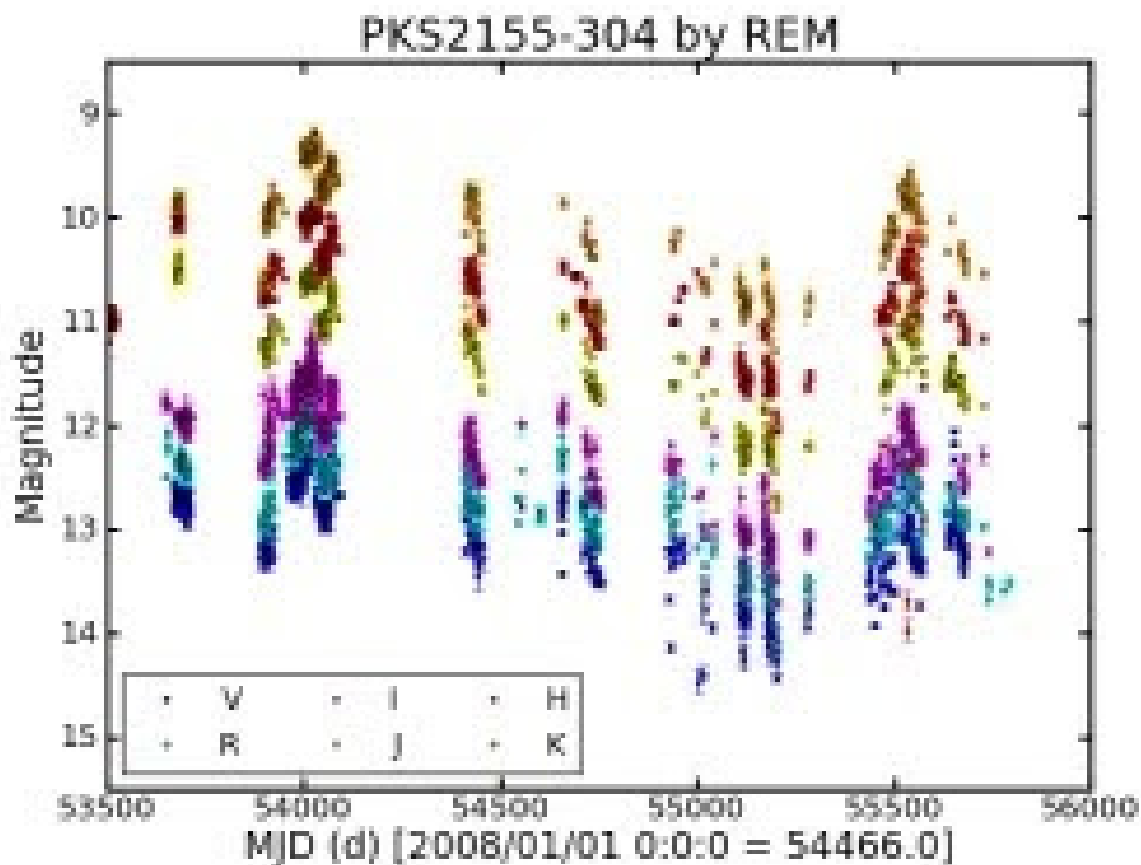
- And now we also have direct imaging of the M87 central black holes:



C. Goddi, Z. Youni, J. Davelaar, M. Kommissar, ESO

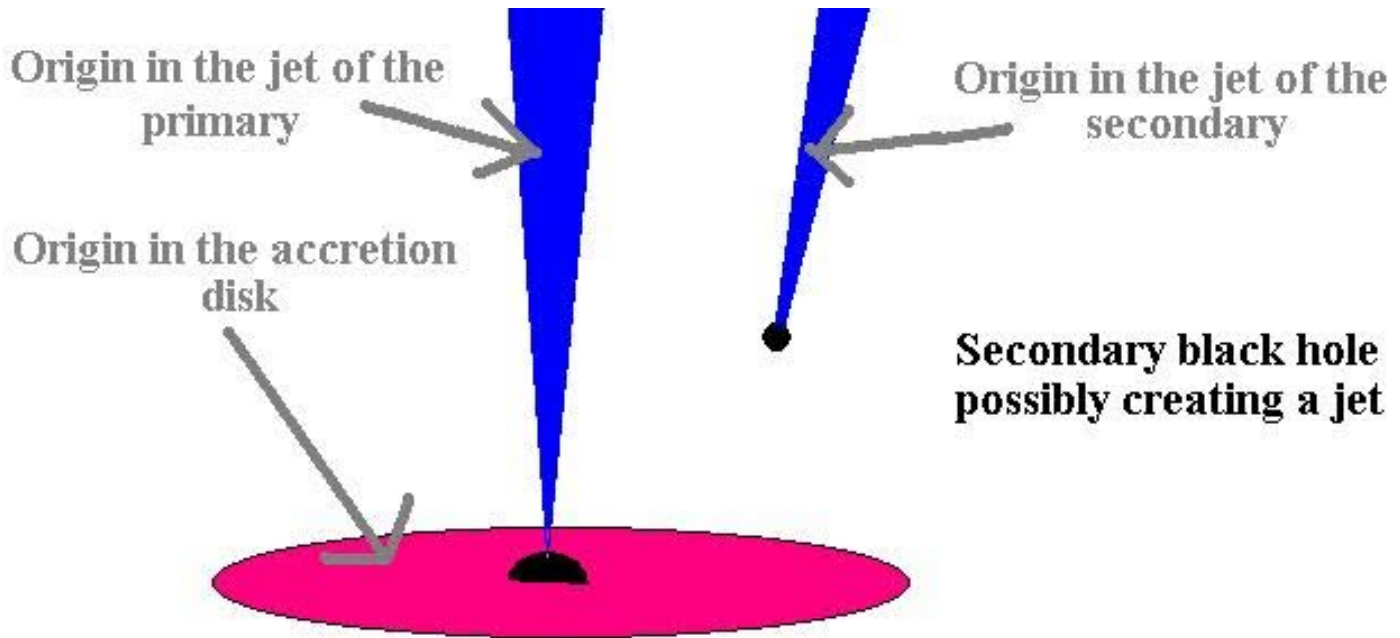
A few more thoughts

- AGN are important for several more reasons:
 - They have produced ~10% of all the luminous energy since the Big Bang.
 - They are unique laboratories for studying physics under extreme conditions.
 - They played a major role in the evolution of the baryonic component of the universe (galaxies and the intergalactic medium).
- Among the various AGN phenomenologies, jets play a fundamental role:
 - If one of the two opposite jets is pointing to the observer (i.e. Earth) we call these sources “blazars”.
 - There are, as usual, subtypes, but in general the jet emission outshines the whole galaxy by a large factor.
 - Plenty of relativistic phenomena to take into account. Blazars dominate the high-energy sky.
- These sources are often monitored since decades with multi-band light curves.
 - This is, for instance, the dataset obtained by the [INAF REM telescope](#) for [PKS2155-304](#):



AGN/Blazar with binary central black-holes?

- There is strong interest in (possible) blazar periodicities.
- It is a highly debated topic, a general agreement in the community is still missing for a series of reasons that we are (likely) able to realize after this exercise:
 - Often, we refer to year-long (quasi-)periodicities, with light-curve at most a couple of decades long. Usual statistical tests are of limited effectiveness.
 - Time-series are affected by red-noise, making evaluation of the significance of identified features difficult.
 - For data from ground-based facilities, sampling is irregular with long (seasonal) gaps.



Primary black hole with accretion disk and jet

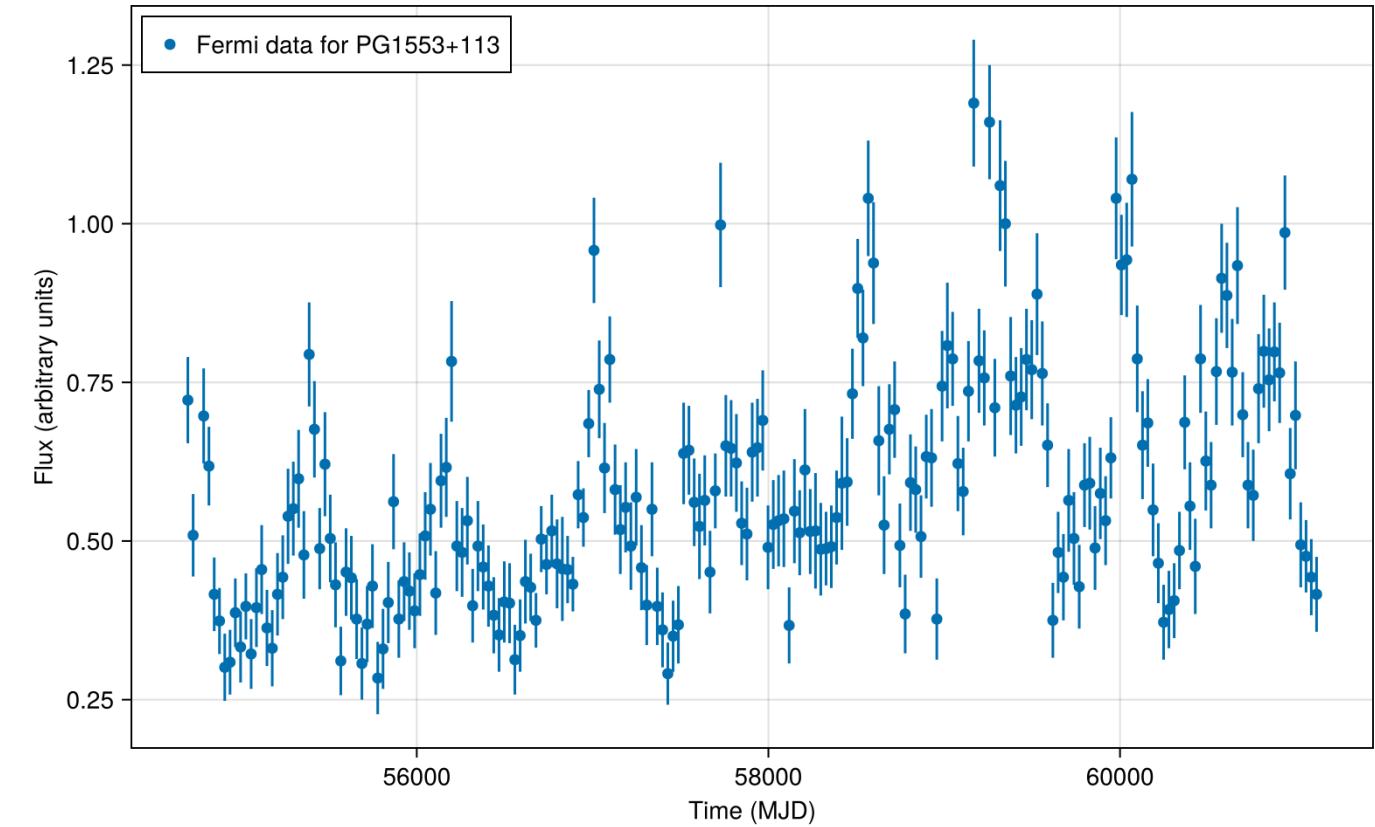
Tip

Possibly the best case identified so far

Exercise: PG1553+113 possible periodicity

- We analyse data obtained by the [Fermi satellite](#) with the [LAT](#) instrument. Data can be downloaded [here](#).
- Let's see the light-curve:

	Date(UTC)	Julian Date	MET	TS	Photon Flux [0.1-100 GeV] (photons cm-2 s-1)	Photon Flux Error(photons cm-2 s-1)	Photon Index	Photon Index Error	
1	"Aug 20 2008"	2.4547e6	240883201	632.49	7.22e-8	6.8e-9	-1.54	"-"	:
2	"Sep 19 2008"	2.45473e6	243475201	366.78	5.09e-8	6.5e-9	-1.54	"-"	:
3	"Nov 18 2008"	2.45479e6	248659201	507.72	6.97e-8	7.5e-9	-1.54	"-"	:
4	"Dec 18 2008"	2.45482e6	251251201	631.78	6.18e-8	6.2e-9	-1.54	"-"	:
5	"Jan 17 2009"	2.45485e6	253843201	269.32	4.16e-8	5.8e-9	-1.54	"-"	:
6	"Feb 16 2009"	2.45488e6	256435201	339.27	3.74e-8	5.2e-9	-1.54	"-"	:
7	"Mar 18 2009"	2.45491e6	259027201	160.2	3.01e-8	5.3e-9	-1.54	"-"	:
8	"Apr 17 2009"	2.45494e6	261619201	170.24	3.09e-8	5.1e-9	-1.54	"-"	:
9	"May 17 2009"	2.45497e6	264211201	295.92	3.87e-8	5.4e-9	-1.54	"-"	:
10	"Jun 16 2009"	2.455e6	266803201	186.56	3.33e-8	5.6e-9	-1.54	"-"	:
altro									
214	"Mar 19 2026"	2.46112e6	795571201	273.94	4.16e-8	5.9e-9	-1.54	"-"	:



- The light-curve shows a possible cyclic behavior. This is indeed one of the sources of this category with the strongest claims for a possible periodicity.
- We first try to model this time-series with an ARIMA model. Data are (approximately) evenly sampled with a point in 30 days.

Mean sample rate: **30.14** days, with standard deviation: **2.06** days.

- Indeed the sampling is pretty regular, but there are a few gaps when source flux was so low not to secure a detection.
- For our purpose here we can assume the sampling is regular.
- Let's first check data stationarity.

Augmented Dickey-Fuller unit root test

Population details:

parameter of interest:	coefficient on lagged non-differenced variable
value under h_0 :	0
point estimate:	-0.25643

Test summary:

outcome with 95% confidence:	reject h_0
p-value:	<1e-04

Details:

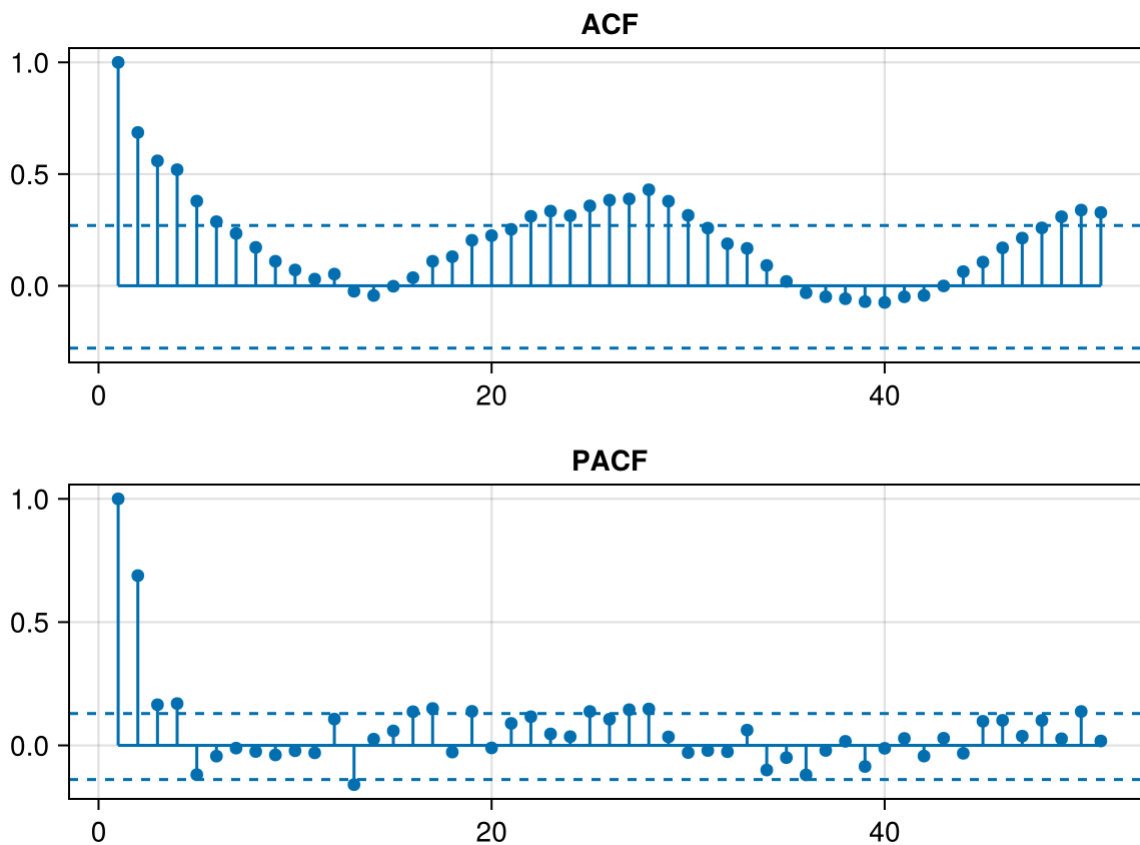
sample size in regression:	212
number of lags:	1
ADF statistic:	-4.77034
Critical values at 1%, 5%, and 10%:	adjoint([-3.46128, -2.87514, -2.57402])

```
1 ADFTest(dt[!,"Rate"],:constant,1)
```

- The light-curve can be considered stationary and therefore we go on modeling it with an ARMA model.
- This essentially means that our null hypothesis is that this time-series can be described by a purely stochastic model.

If this were the case, the light-curve, being only due to stochastic processes, has no significant QPOs.

- First, let's look at the ACF and PACF diagnostic plots.



- The ACF is zero after the first 7 lags, but it also shows the typical cyclic behavior of periodic phenomena, with a period of about 25 lags ~ 2 years that is close to the proposed (quasi-)periodicity of 2.2 years (Achermann et al. 2015).
- The PACF is zero after 3 lags.
- However, due to the possible QPO, the diagnostic plots are not easy to interpret and we prefer choosing the AR and MA orders by modern computational tools carrying out an optimized search.

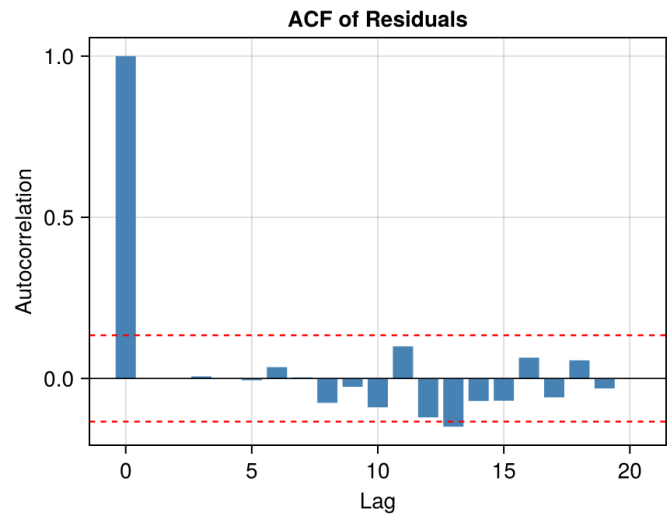
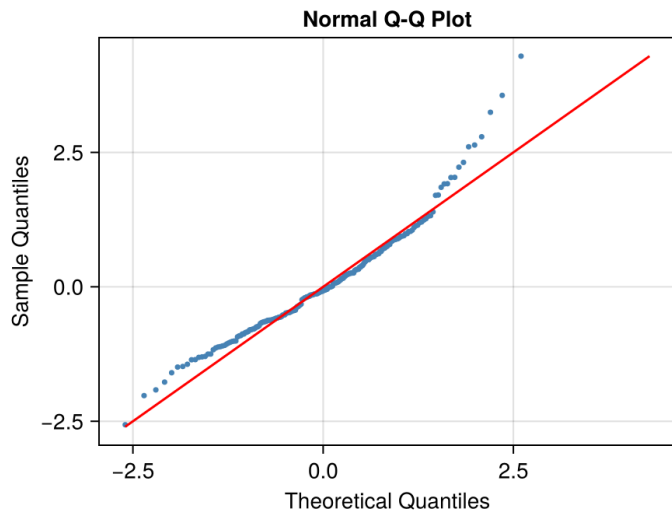
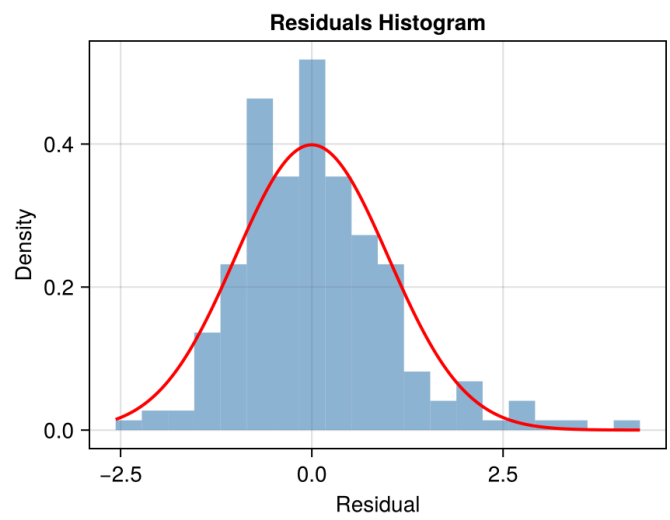
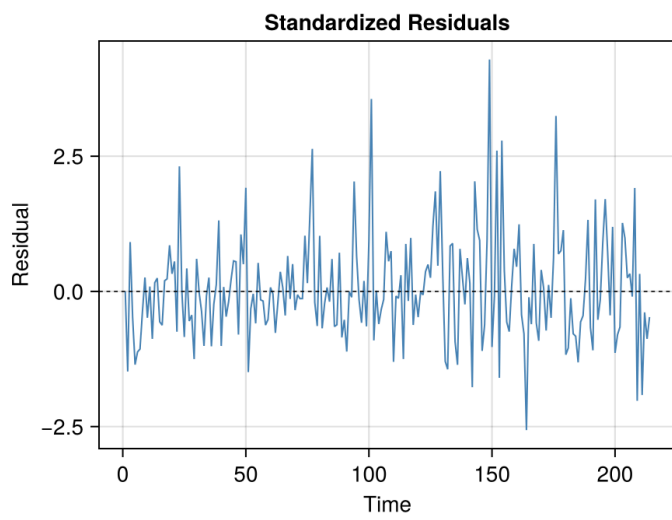
```
1 amodel = auto_arima(dt[!,:Rate],allow_mean=true);
```

The best selected ARMA components turns out: AR(1), MA(4) with 1 differentiations. No seasonal components are selected.

- Results are indeed not so different from those inferred from the ACF/PACF analysis. Let's see some diagnostic plots.

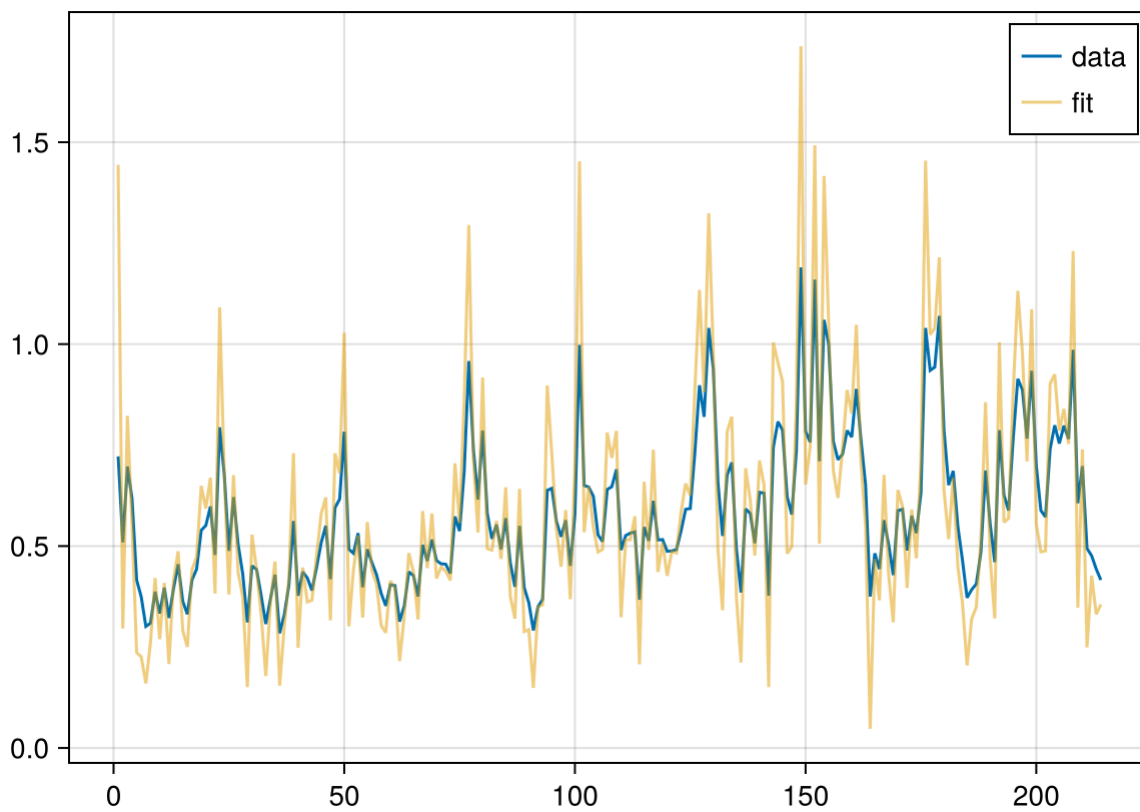
```
1 kf = kalman_filter(amodel);
```

```
1 begin
2   # --- Estrai i residui standardizzati ---
3   residuals = [kf.v[t][1] / sqrt(kf.F[t][1,1]) for t in 1:length(kf.v)]
4   n = length(residuals)
5   t = 1:n
6
7   # --- Calcola ACF dei residui ---
8   max_lag = min(20, n ÷ 4)
9   acf_vals = autocor(residuals, 0:max_lag)
10  conf_bound = 1.96 / sqrt(n) # banda di confidenza 95%
11 end;
```



- And we immediately see that we have discrepancies in particular for low and the high fluxes, i.e. the variations are too high for a pure stochastic phenomenon.
- This might also be due to the hint of periodicity that cannot be described by a pure stochastic model as the ARMA process we are testing.

ARMA fit



- What (provably) happens is simply that the model, in order to reproduce the positive spikes often interpreted as due to a QPO as due to a pure stochastic process, needs to have almost equally likely positive and negative variations we indeed do not see in our data.

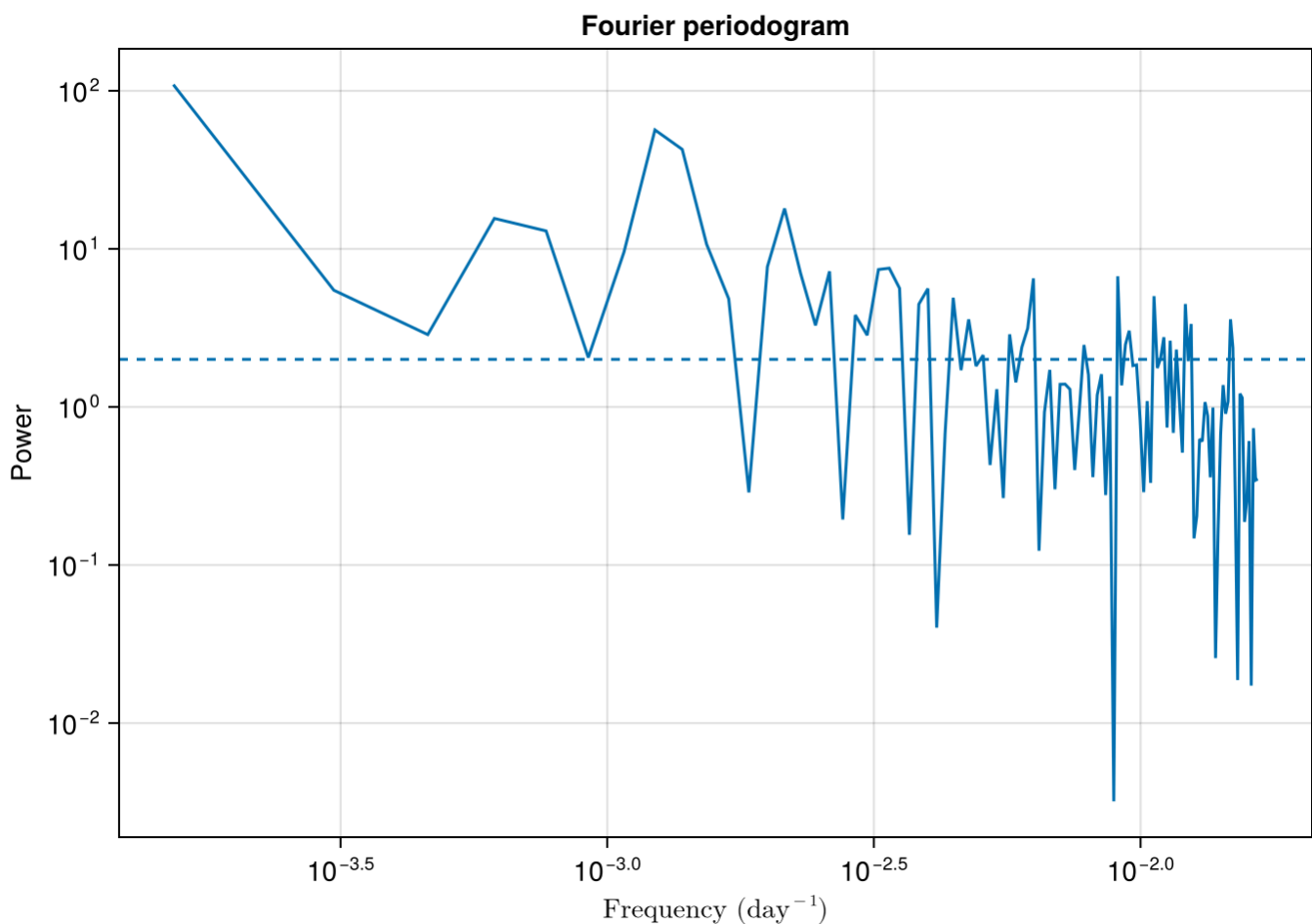
Tip

Let's try a pure Bayesian approach modeling the PDS of the source

- This is a fairly sophisticated approach, based on a procedure described by [Vaughan \(2010\)](#).
- Let's compute the Nyquist frequency and a lower-limit for the frequencies accessible with our dataset:

Nyquist frequency: 0.0166 days^{-1} , lowest frequency: 0.0002 days^{-1}

- And let's compute the periodogram indicating the expected Poissonian noise. The (large) peak at ~ 2.2 years is clearly visible.



- We plan to model the noise of the time-series as PL plus a constant, i.e. the Poissonian noise level. The PL+constant model is the null hypothesis of a time-series generated by noise only, i.e. without any real periodicity.

model (generic function with 1 method)

```
1 function model(theta, x)
2     lnn,a,g = theta
3     return exp.(lnn) .* x.^a .+ abs.(g)
4 end
```

- We are going to carry out a Bayesian analysis and we need to define a proper likelihood. In this case, due to the statistics affecting a periodogram, we work with the Whittle likelihood.

lnlike (generic function with 1 method)

```
1 function lnlike(theta, x, y)
2     lnn,a,g = theta
3     return -2.0 .* sum(y ./ model(theta,x) .+ log(model(theta,x)))
4 end
```

- And define a probabilistic model for the noise.

permodel (generic function with 2 methods)

```
1 @model function permodel(x,y)
2     #
3     lnn ~ Uniform(-15,0)
4     a ~ Truncated(Normal(0.,10.),-10,0)
5     g ~ Normal(2,0.4)
6     #
7     Turing.@addlogprob! lnlike((lnn,a,g),x,y)
8     #
9     for i in 1:length(y)
10         y[i] ~ Chisq(2) * model((lnn,a,g),x[i]) / 2
11     end
12     #
13 end
```

- Now, let's prepare and run the MCMC chain.

	iteration	chain	lnn	a	g	n_steps	is_accept	acceptance_rate	alt
1	1001	1	-6.74497	-1.39191	0.692312	15.0	1.0	0.886206	
2	1002	1	-6.69365	-1.36716	0.771999	15.0	1.0	1.0	
3	1003	1	-6.68741	-1.37108	0.643571	15.0	1.0	0.910356	
4	1004	1	-6.0445	-1.26536	0.66726	31.0	1.0	0.595711	
5	1005	1	-6.14379	-1.26891	0.686916	15.0	1.0	0.454578	
6	1006	1	-6.09948	-1.27804	0.689783	5.0	1.0	0.787298	
7	1007	1	-5.73771	-1.22206	0.495982	41.0	1.0	0.991777	
8	1008	1	-6.06739	-1.26748	0.632645	31.0	1.0	0.982015	
9	1009	1	-5.24701	-1.14696	0.502579	23.0	1.0	0.489173	
10	1010	1	-6.00094	-1.2451	0.573733	31.0	1.0	0.73706	
	altro								

```
1 begin
2   iterations = length(freq)*250
3   burnins = 1000
4   chains = 4
5
6   chain = mapreduce(c -> sample(permodel(freq,power), NUTS(burnins,0.65),
7     iterations), chainscat, 1:chains)
7 end
```

Sampling

Found initial step size
ε: 0.025

Sampling

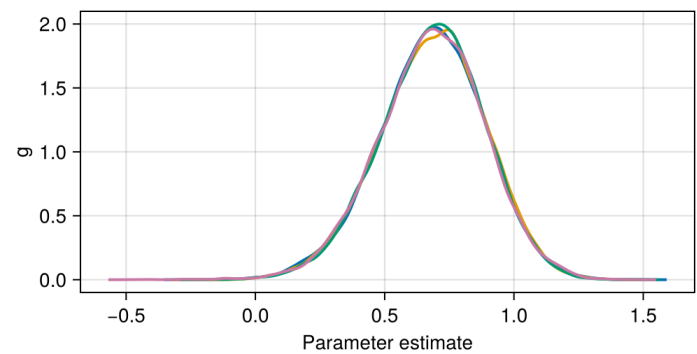
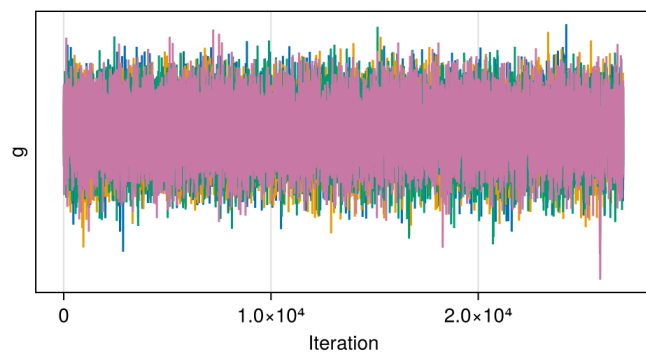
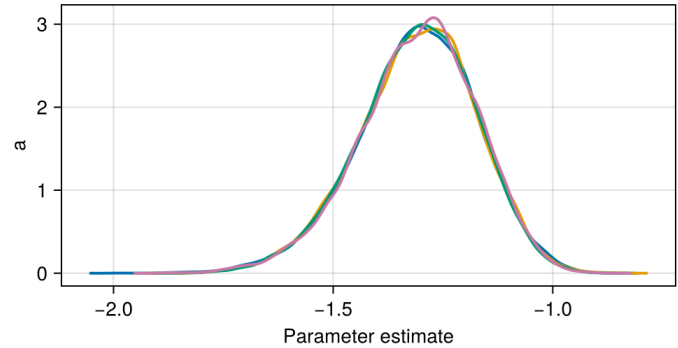
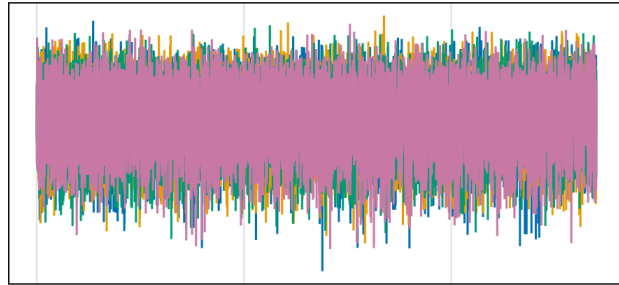
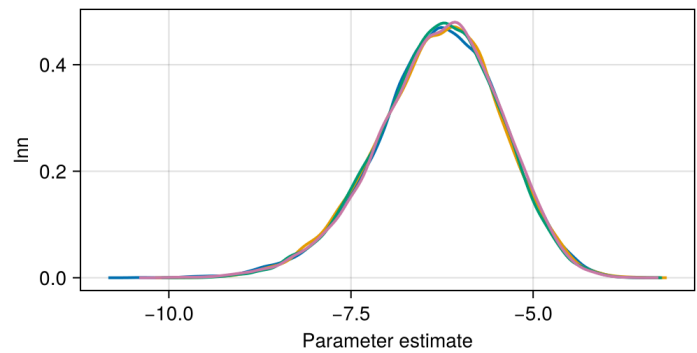
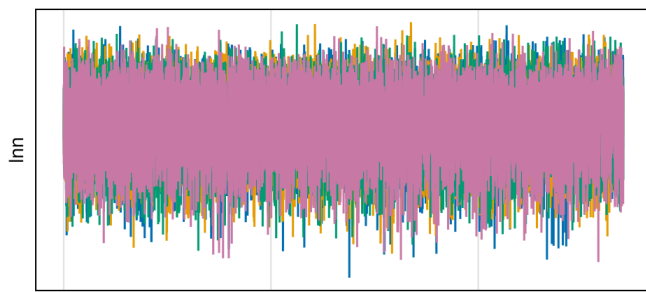
Found initial step size
ε: 0.003125

Sampling

Found initial step size
ε: 0.025

Sampling

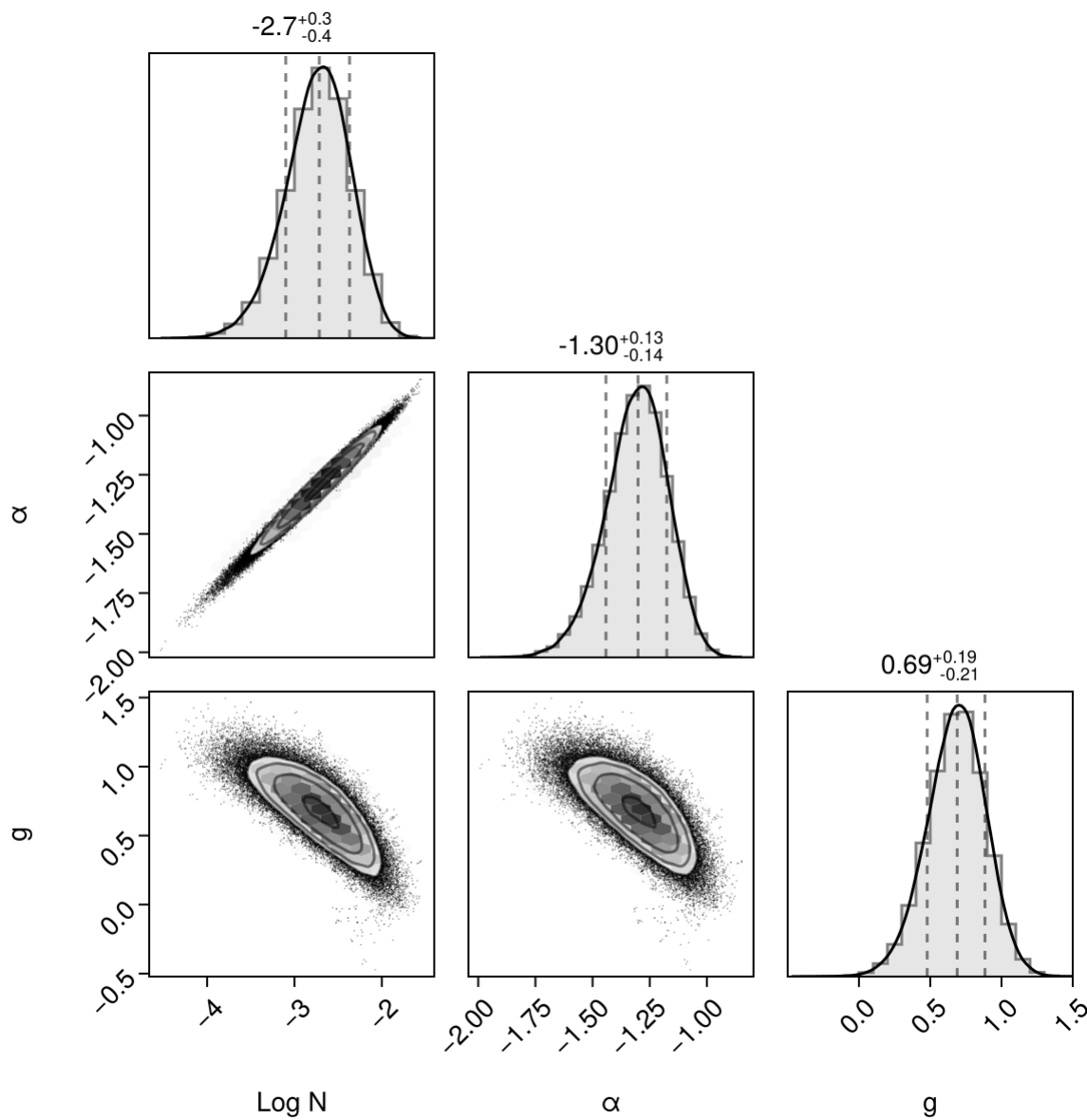
Found initial step size
ε: 0.025



- Checking chain convergence is very important.
- The $\hat{r}$ parameter, or Potential Scale Reduction Factor (PSRF), is a MCMC convergence diagnostic that compares between-chain variance to within-chain variance for multiple chains.
- It measures if chains have mixed well; a value of $\hat{r} \approx 1$ indicates convergence.
- It has been introduced in [Gelman & Rubin \(1992\)](#).

	parameters	rhat
1	:lnn	1.00014
2	:a	1.00013
3	:g	1.00016

- The chains look fine.
- We can also check the chain result by a classic *corner plot*!



- The posteriors are well defined and single-peaked.
- Since the statistics is χ^2 with 2 degrees of freedom we can check whether the model is properly fitting the data, as it seems to be the case.

Exact one sample Kolmogorov-Smirnov test

Population details:

parameter of interest:	Supremum of CDF differences
value under h_0 :	0.0
point estimate:	0.0956652

Test summary:

outcome with 95% confidence:	fail to reject h_0
two-sided p-value:	0.2593

Details:

number of observations:	108
-------------------------	-----

```

1 begin
2   theta = [mean(chain["lnn"]),mean(chain["a"]),mean(chain["g"])]
3
4   ExactOneSampleKSTest(2*power./model(theta,freq), Chisq(2))
5 end

```

- Our test is based on the fact that the adopted likelihood implies that the ratio of the periodogram ordinates, the powers, I_j to the *true* spectrum S_j will follow a χ^2 distribution with 2 degrees of

freedom.

$$R_j^{obs} = 2I_j^{obs}/S_j(\theta)$$

- Where $S_j(\theta)$ is a parametric spectral model with known parameters.
- Details of the procedure we follow are discussed in [Vaughan \(2010\)](#).
- Now, let's compute, basing upon the MSMC chains, the percentile for any chosen level for each frequency bin.

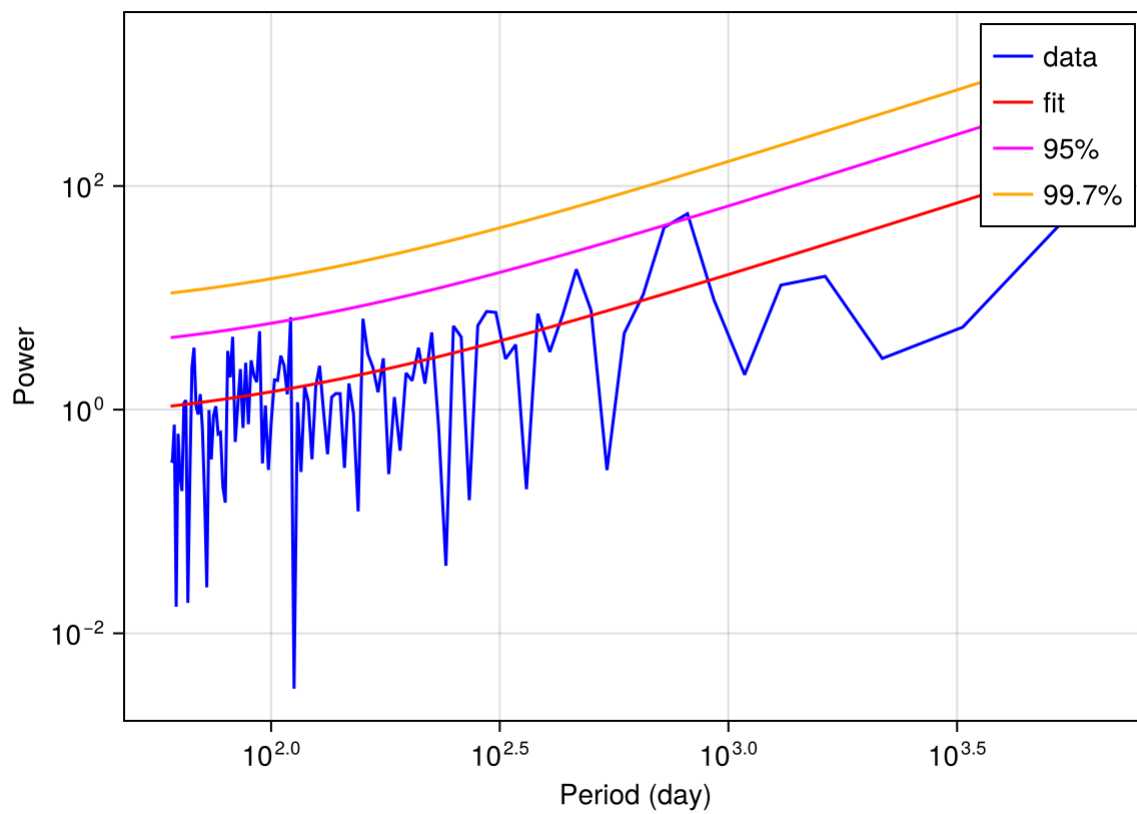
```
1 begin
2     # Compute the best fit model
3     Sbest = model(theta,freq)
4
5     # Derive a set of models based on the posterior distribution
6     S = [model((dfc[i,:lnn],dfc[i,:a],dfc[i,:g]),freq) for i in 1:nrow(dfc)]
7
8     # Multiply a chi2 distribution to the models and derive the simulated periodograms
9     I = S .* rand(Chisq(2),length(S))/2.
10
11    # Compute the observed
12    Robs = 2*power./Sbest
13 end;
```

```
1 begin
2     # Compute the simulated Rs
3     Is = reshape(I,(Int(length(I)/length(Sbest)),length(Sbest)))
4     Rsim = [2*Is[i,:][1] ./ Sbest for i in 1:size(Is)[1]]
5 end;
```

```
1 begin
2     # And derive any chosen percentile on the simulated distribution
3     R95 = percentile([maximum(Rsim[i]) for i in 1:length(Rsim)],95.0)
4     R997 = percentile([maximum(Rsim[i]) for i in 1:length(Rsim)],99.7)
5
6     # And convert back to periodogram values.
7     pp95 = R95.*Sbest/2.
8     pp997 = R997.*Sbest/2.
9 end;
```

- And finally let's plot our periodogram together with the predictions for pure noise datasets.

PG1553+113



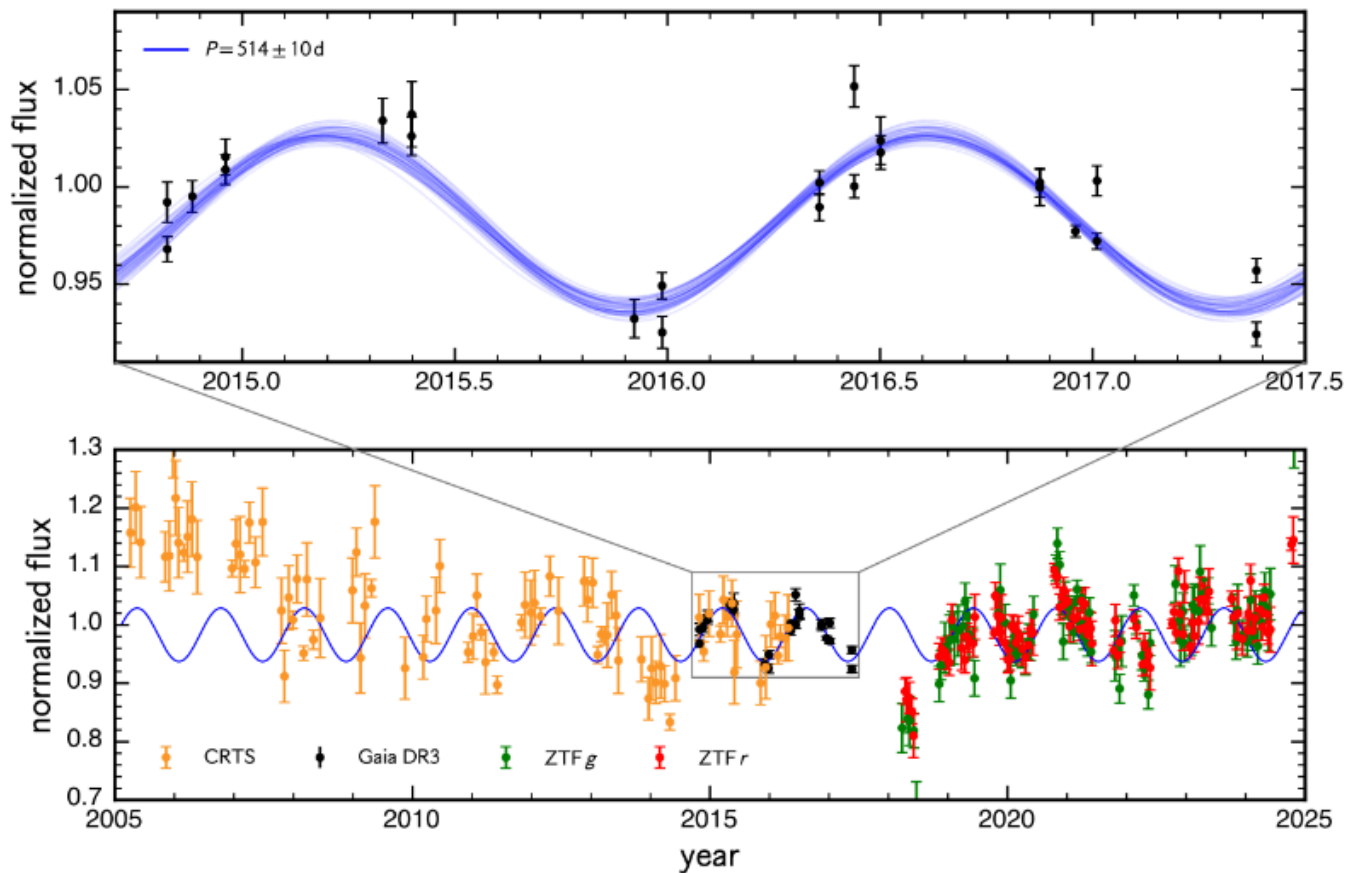
- The peak in the periodogram at ~ 2.2 years, although of interest, does not reach the 3σ limit threshold, and we cannot rule out the possibility it is a noise artifact.

Please, be aware that in the literature there is a very hot debate about a possible QPO in this source!

- It is true that this is anyway the most promising case in this category.

A word of caution

- As already mentioned, finding possible QPOs in AGN is a very hot topic, but the analysis is also of great complexity and the rate of false positive is almost always underestimated for various reasons, e.g. too simple noise modeling, ineffective "look elsewhere effect" correction, inadequate analysis tools, etc.
- In the following figure, taken from [El-Badry et al. \(2025\)](#), we see the light-curve for an AGN obtained by the [Gaia](#) satellite, and by a few more ground based facilities.



- If the analysis is based on Gaia data only, a clear and sinusoidal periodicity appears that was considered significant, although based on only a couple of cycles. However, enlarging the monitoring length with data from different sources it is clear that there is no stable periodicity. The recurrent pattern turns out an artifact, likely introduced by the correlated noise.

Reference & Material

Material and papers related to the topics discussed in this lecture.

- [Covino et al. \(2019\)](#) - "Gamma-ray quasi-periodicities of blazars. A cautious approach"
- [El-Badry et al. \(2025\)](#) - "Active galactic nuclei do not exhibit stably periodic brightness variations"

Further Material

Papers for examining more closely some of the discussed topics.

- [Ackermann et al. \(2015\)](#) - "Multiwavelength Evidence for Quasi-periodic Modulation in the Gamma-Ray Blazar PG 1553+113"
- [Vaughan \(2010\)](#) - "A Bayesian test for periodic signals in red noise"

Credits

Course Flow

	Previous lecture	Next lecture	Course Summary
notebook	Lecture about time domain analysis	Lecture about time of arrival	Course Summary
html	Lecture about time domain analysis	Lecture about time of arrival	Course Summary

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